INCH-POUND
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PERFORMANCE SPECIFICATION

RUBBER OR PLASTIC SHEETS AND ASSEMBLED AND MOLDED SHAPES, SYNTHETIC, FOAM OR SPONGE, OPEN CELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers inherently flame-resistant flexible foam or sponge materials with low emission of smoke and toxic products to be used for cushioning applications, such as upholstered furniture, and for packaging applications. The foam will consist of synthetic rubber or plastic synthetic material and will be open cell. The foam will be assembled in the form of sheets or molded shapes.

1.2 Classification. The foam will be available in the following conditions (see 6.2), which reflect the firmness rating of the foam:

a. Condition A – Medium
b. Condition B – Medium firm
c. Condition C – Firm
d. Condition D – Soft

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

CCC-C-436 - Cloth, Ticking Twill, Cotton

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-293 - Visual Inspection Guide for Cellular Rubber Items

MIL-STD-1623 - Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use)

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual

(Copies of the chapter titled "Material Control Program" are available by email request to CommandStandards@navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM C739 - Standard Specification for Cellulosic Fiber Loose-Fill Thermal Insulation

ASTM D573 - Standard Test Method for Rubber–Deterioration in an Air Oven

ASTM D3574 - Standard Test Methods for Flexible Cellular Materials–Slab, Bonded, and Molded Urethane Foams

ASTM E162 - Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source

ASTM E662 - Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials

ASTM E800 - Standard Guide for Measurement of Gases Present or Generated During Fires

ASTM E1590 - Standard Test Method for Fire Testing of Mattresses

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. The base material shall be a synthetic rubber or plastic cellular compound processed to meet all of the requirements of this specification. The use of reclaimed or natural rubber is prohibited (see 6.2).

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction.

3.3.1 Form. The material shall have a polymerized or vulcanized homogeneous (free from foreign material) cellular structure with a porous surface and open cells. The cells shall be interconnecting and uniform in size. The form of the material shall be in sheets or in assembled or molded shapes, as specified (see 6.2). Solid fillers, flame-retardant chemicals, and other additives used to achieve appropriate performance standards as specified in this specification shall not be deemed “foreign” material within the meaning of this section.

3.3.2 Manufacturing requirements. Cellular material may be molded in one piece or may be assembled by laminating to achieve thickness and by butting to achieve width or length. Assembled items shall be cemented together. Incidental defects in the cellular material, which occur in standard manufacturing practices and assembly, shall be allowed if they are corrected as indicated below and the resulting product conforms to this specification:

a. Tears occurring during removal from the mold shall be cemented together.

b. Changes in volume during cure, resulting in surface depressions or minor changes in size, shall be corrected by the addition of material of identical composition and buffed to a level surface.

c. Laminae products may be scrim-reinforced provided such scrim-reinforcing complies with flammability and smoke emission requirements.

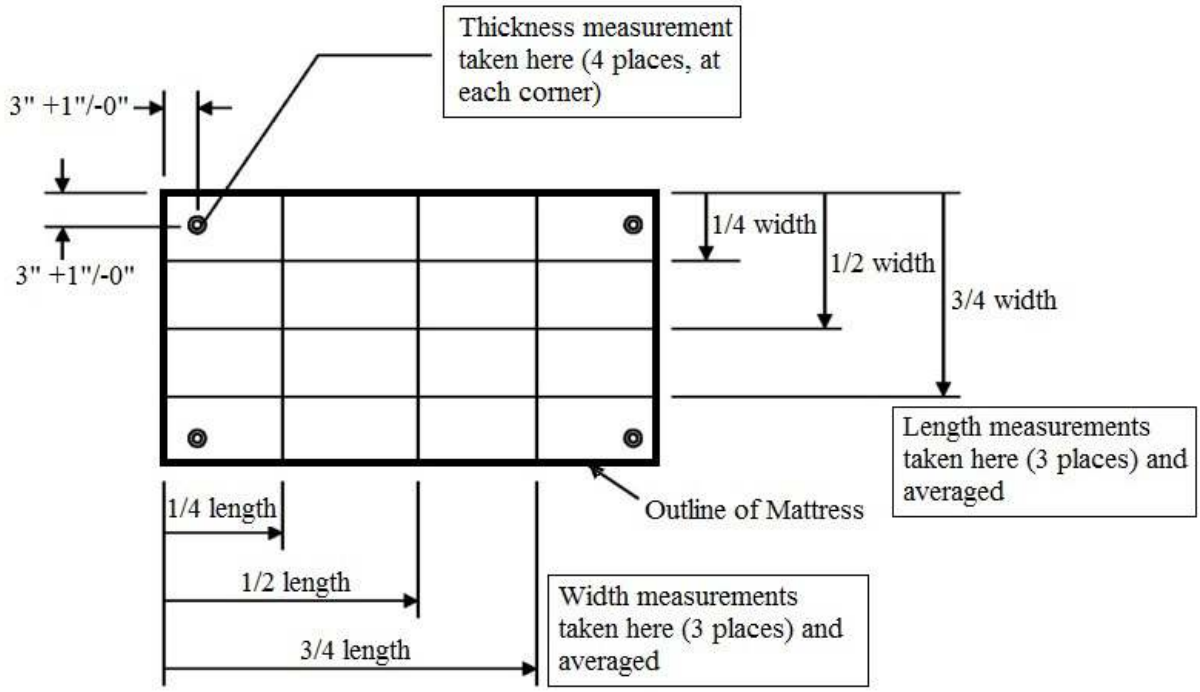
d. Where laminae assembly is employed to achieve thickness, the laminae shall be as thick as possible but in no case less than 0.5 inch.

e. A maximum of two sections may be butted together to achieve required width or length. Sections are defined as pieces of full thickness that are the full width or length of the finished foam. Lamination may be used to achieve thickness. No more than one butted joint is allowed in any single lamination.

3.4 Odor. The cellular material shall be properly cured in order to prevent any objectionable odor. An objectionable odor is defined as one that is atypical of a cellular rubber compound. The material shall be in accordance with the odor emission requirements of ASTM C739.

3.5 Dimensions and tolerances. The dimensions of all sheets or molded or assembled shapes shall be as specified (see 6.2). For sheets longer than 2 feet in length, the dimensions of the test specimen shall be measured at the locations in accordance with guidance provided on figure 1.

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NOTE:

1. The tolerance for locations of the length and width measurements is ±1 inch.

FIGURE 1. Test specimen dimensional measurement locations.

3.6 Physical properties.

3.6.1 Indentation. The indentation of each condition of cellular material shall be as specified in table I.

TABLE I. Indentation and apparent density of cellular cushioning material.

Condition	Indentation Force Deflection, lbf [1/], required to compress 50 in^2 [2/] of a 3-inch specimen 25% in accordance with ASTM D3574	Maximum apparent density, lb/ft^3 [3/], based on gross volume
Medium	25 to 42	8.0
Medium firm	30 to 55	9.0
Firm	42 to 60	9.2
Soft	15 to 30	7.5

NOTES:
[1/] Pounds-force.
[2/] Square inches.
[3/] Pounds per cubic foot.

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3.6.2 Density. The maximum apparent density of each condition of cellular material shall be as specified in table I.

3.6.3 Compression set. The compression set of the cellular material shall be not greater than 12 percent.

3.6.4 Fire performance.

3.6.4.1 Flame spread (radiant panel) index. The cellular material shall demonstrate a flame spread index no greater than that specified in MIL-STD-1623 when tested in accordance with 4.7.4.2. The specimen shall not exhibit melting or flaming drops.

3.6.4.2 Smoke density. The cellular material shall demonstrate maximum specific optical density of smoke no greater than that specified in MIL-STD-1623 when tested in accordance with 4.7.4.3. The specimen shall not melt or drip.

3.6.4.3 Fire gas toxicity. The finished foam material shall be evaluated for types of gases and the toxicity levels released in accordance with 4.7.4.4.

3.6.4.4 Open flame test.

3.6.4.4.1 Heat release rate. The cellular material shall have a maximum net peak heat release rate not greater than 150 kilowatts (kW) when tested in accordance with 4.7.4.5.

3.6.4.4.2 Flaming material. No flaming droplets or flaming material is allowed.

3.6.4.4.3 Smoke generation. The cellular material shall have an average specific extinction area not greater than 1,468 square feet per pound (ft^2/lb) (300 square meters per kilogram [m^2/kg]) when tested in accordance with 4.7.4.5.

3.6.5 Bend fatigue resistance. The cellular material shall not crack before or after aging when tested in accordance with 4.7.5.

3.6.6 Tensile strength and ultimate elongation. The tensile strength of the material shall be not less than 9 pounds per square inch (lb/in^2), and ultimate elongation shall be not less than 100 percent when tested in accordance with 4.7.6.

3.6.7 Cemented seam strength. The cemented seams shall be stronger than the base material, both initially and after oven aging, when tested in accordance with 4.7.7.

3.6.8 Dynamic fatigue. The material thickness shall not decrease by more than 10 percent maximum when tested in accordance with 4.7.9.

3.7 Identification. Each sheet or molded or assembled shape shall be marked in a permanent and legible manner with the following information. If the item is too small for marking, the information shall be provided on the wrapping or container of each item (see 6.2).

a. Manufacturer
b. Date of manufacture
c. Contract number and lot (if applicable)
d. Specification number
e. Condition

3.8 Workmanship. The material shall be of uniform texture, be free of obvious manufacturing imperfections that may impair serviceability, and meet the requirements of this specification.

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3.9 Off-gassing. For materials used in submarines (see 6.2), the finished foam shall be evaluated for off-gassing in accordance with the requirements of 4.8. Based on the circumstances of use and the chemical nature of the finished foam, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the finished foam shall be assigned to either the "Permitted" or "Limited" usage category (see 4.8 and 6.4).

3.10 Toxicity and prohibited materials.

3.10.1 Toxicity. When evaluated in accordance with 4.9, the finished foam shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.9 and 6.6).

3.10.2 Prohibited materials. The finished foam shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the end item when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 and the tests specified in table II.

4.2.1 Sampling for first article inspection. Sample sheets or molded or assembled shapes shall be randomly selected and tested in accordance with 4.2. The rate of sampling for examination shall be 3 units on lots of 250 or less, 5 units on lots of 251 to 1,200, 7 units on lots of 1,201 to 10,000, and 10 units on lots of 10,001 to 35,000.

4.2.2 Toxicity testing. Toxicity shall be evaluated in accordance with 3.10 and 4.9.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests specified in table II.

4.3.1 Sampling for conformance inspection. Sample sheets or molded or assembled shapes shall be randomly selected from each lot and tested in accordance with 4.3. The rate of sampling for examination shall be 3 units on lots of 250 or less, 5 units on lots of 251 to 1,200, 7 units on lots of 1,201 to 10,000, and 10 units on lots of 10,001 to 35,000.

4.4 Lot. A lot shall consist of all materials of the same condition produced in one plant under the same conditions and offered for delivery at one time.

4.5 Examination. Each of the samples taken in accordance with 4.5.1 shall be examined for workmanship (see 3.8), construction and visual defects (see 3.3), dimensions (see 3.5), tolerances (see 3.5), and marking (see 3.7). Determination and evaluation of visual defects shall be in accordance with MIL-STD-293. Major and minor defects shall be classified in accordance with MIL-STD-293.

4.5.1 Sampling for examination. Sample sheets or molded or assembled shapes shall be randomly selected from each lot. The rate of sampling for examination shall be 3 units on lots of 250 or less, 5 units on lots of 251 to 1,200, 7 units on lots of 1,201 to 10,000, and 10 units on lots of 10,001 to 35,000.

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TABLE II. First article and conformance inspection.

Characteristic	Requirement	Test procedure	First article inspection	Conformance inspection	Number of determinations per sample unit	Results reported as
Material	3.2	4.10	X	-	1	Pass or fail
Construction	3.3.1, 3.3.2.a, b, d, and e	4.5	X	X	1	Pass or fail
Odor	3.4	4.7.8	X	-	5	Required by ASTM C739
Indentation	3.6.1	4.7.1	X	X	1	To nearest 0.5 (lbf) per 50 in^2
Density	3.6.2	4.7.2	X	X	1	To nearest 0.1 lb/ft^3
Compression set	3.6.3	4.7.3	X	X	3	To nearest 0.1%
Flame spread index	3.6.4.1	4.7.4.2	X	X	4	Pass or fail
Smoke density	3.3.2.c, 3.6.4.2	4.7.4.3	X	X	6	Pass or fail
Fire gas toxicity	3.6.4.3	4.7.4.4	X	X	4	Parts per million
Open flame test	3.6.4.4	4.7.4.5	X	X [1/]	1	Pass or fail
Bend fatigue:						
Before aging	3.6.5	4.7.5	X	-	1	Pass or fail
After aging	3.6.5	4.7.5	X	-	1	Pass or fail
Tensile strength	3.6.6	4.7.6	X	X	3	To nearest 0.1 lb/in^2
Ultimate elongation	3.6.6	4.7.6	X	X	3	To nearest 1%
Cemented seam strength:						
Before aging	3.6.7	4.7.7	X	-	1	Pass or fail
After aging	3.6.7	4.7.7	X	-	1	Pass or fail
Dynamic fatigue	3.6.8	4.7.9	X	-	1	To nearest 1%
Off-gassing	3.9	4.8	X	-	1	Permitted or Limited
Toxicity	3.10	4.9	X	-	1	Pass or Fail

NOTE:
[1/] Conformance testing required only for specimens that would be used as mattresses in end item use.

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4.6 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the temperature and relative humidity conditions specified in the relevant test standards. Specimens shall be conditioned in an atmosphere of 50±5 percent relative humidity and a temperature of 70±5 °F.

4.7 Test methods. Testing of the completely fabricated item shall be as specified in table II for the characteristics shown therein. Tests shall be conducted in accordance with applicable methods as specified herein.

4.7.1 Indentation. The indentation of cellular materials shall be determined in accordance with ASTM D3574 Test B_1, except that test specimens shall be not less than 15 by 15 by 3 inches.

4.7.2 Density. The density or apparent density shall be determined in accordance with ASTM D3574.

4.7.3 Compression set. The compression set of the cellular material shall be determined in accordance with the constant deflection compression test set under the constant deflection procedure of ASTM D3574. Specimens shall be compressed to 50 percent of their original height. The compression set shall be calculated and expressed as a percentage of the original height of the sample tested.

4.7.4 Fire performance.

4.7.4.1 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the NAVSEA Technical Authority. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) or International Code Council's International Accreditation Services (IAS). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.7).

4.7.4.2 Flame spread (radiant panel) index. The cellular material shall be tested in accordance with the test procedure specified in ASTM E162. The size of the test specimen shall be 18 by 6 inches by 1 inch thick.

4.7.4.3 Smoke density. The cellular material shall be tested in accordance with the test procedure specified in ASTM E662. The size of the test specimen shall be 3 by 3 inches by 1 inch thick. The tests shall be conducted in both flaming and non-flaming modes.

4.7.4.4 Fire gas toxicity. During the conduct of the smoke density test specified in 3.6.4.3, the finished foam material shall be evaluated for fire gas toxicity in accordance with ASTM E800. The average value of the maximum value of the gas concentration measured at each condition (flaming and non-flaming) shall be recorded for the following gases, as specified (see 6.2) : carbon monoxide (CO), hydrogen chloride (HCl), hydrogen fluoride (HF), nitrogen oxides (NOx), hydrogen bromide (HBr), hydrogen cyanide (HCN), and sulfur dioxide (SO_2). When making toxicity measurements, the sampling of gases shall be made during the testing of the second and the third specimen at each test condition from the geometrical center of the chamber at the time when the maximum specific optical density of smoke is reached. The number of samples shall be four; two each in the flaming and non-flaming condition. The concentration of each toxic gas shall be determined as parts per million (ppm) in the chamber volume.

4.7.4.5 Open flame test. The cellular material to be tested shall have the following dimensions: 26 by 76 inches by 4 inches thick. For the purpose of this testing, the cellular material shall be wrapped in a cotton ticking cover. The ticking shall be in accordance with CCC-C-436, Type II, Class 2. The seams shall be closed on all sides and the cover shall be a snug fit. The open flame test shall be conducted in accordance with ASTM E1590, including all optional results, with the exceptions noted in 4.7.4.5.1. Any failure shall be cause for rejection of the lot.

4.7.4.5.1 Test exceptions. The cellular material shall be tested in accordance with the open flame test contained in 4.7.4.5 with the following exceptions:

a. The test sample shall be mounted label side up on an open frame bunk or support system that will allow direct flame impingement by the ignition source to the bottom surface and sides of the cellular material covered in cotton ticking (see 4.7.4.5). The welded test support frame shall be in accordance with figure 2.

b. ASTM E1590 Test Configuration C shall be used.

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c. The “T” burner described in the Ignition Source section of ASTM E1590 shall be replaced with a burner as shown on figure 3, having a nominal 12- by 12-inch top surface. The burner shall have a minimum 4-inch layer of silica sand and be 0.08 inch (2 millimeters) to 0.12 inch (3 millimeters) in size to provide the horizontal surface through which the gas is supplied. The sand layer shall be flush with the top of the burner.

d. The burner shall be positioned such that the top edge of the burner is 15±0.5 inches below the bottom edge of the test specimen and centered with respect to the long side of the test specimen with one-half of the width of the burner under the cushioning test specimen as shown on figure 4.

e. The test specimen shall be exposed to a fire producing a total heat output of 50±3 kilowatts for the first 5 minutes of the test, followed by an increase to 100±5 kilowatts for an additional 10 minutes. The total fire exposure time shall be 15 minutes.

f. Measurements for carbon monoxide (CO) and carbon dioxide (CO_2) shall be recorded.

g. The burner shall be turned off. Combustion shall be allowed to continue until one or more of the following conditions are reached:

(1) All flaming combustion has ceased.

(2) 30 minutes have elapsed from the time the burner was ignited.

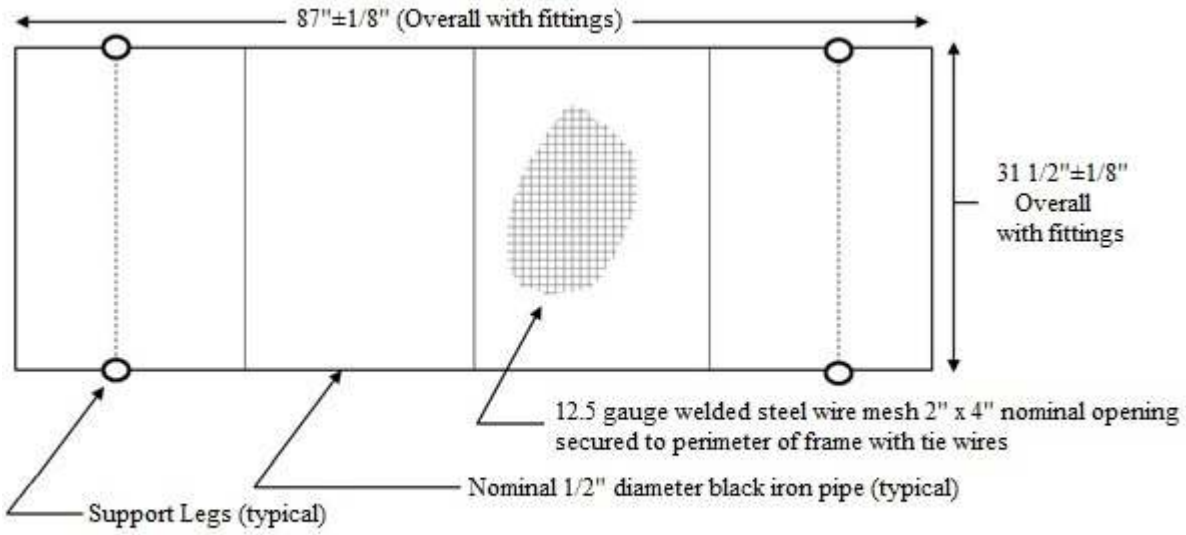


FIGURE 2. Test specimen bed frame.

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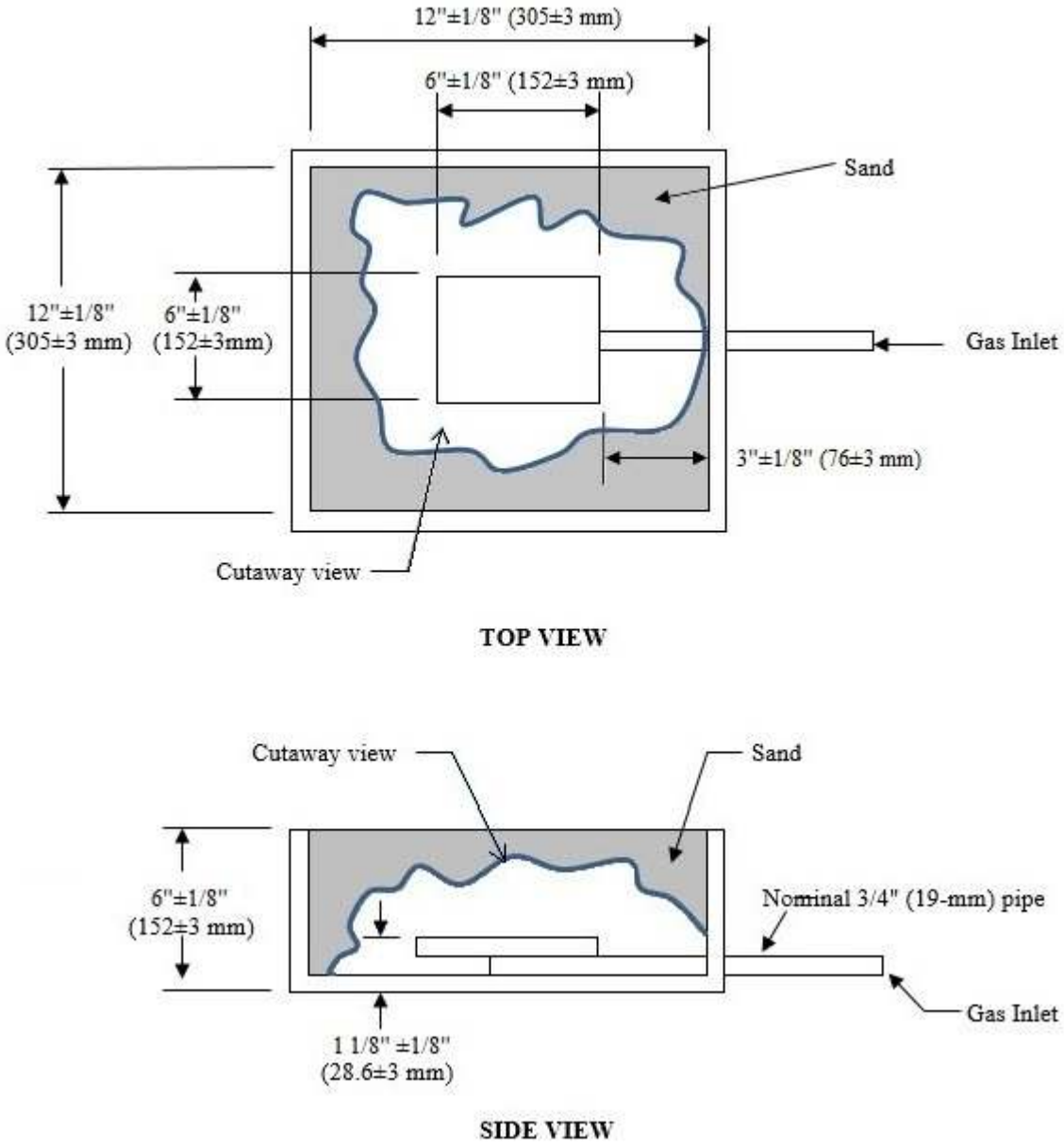
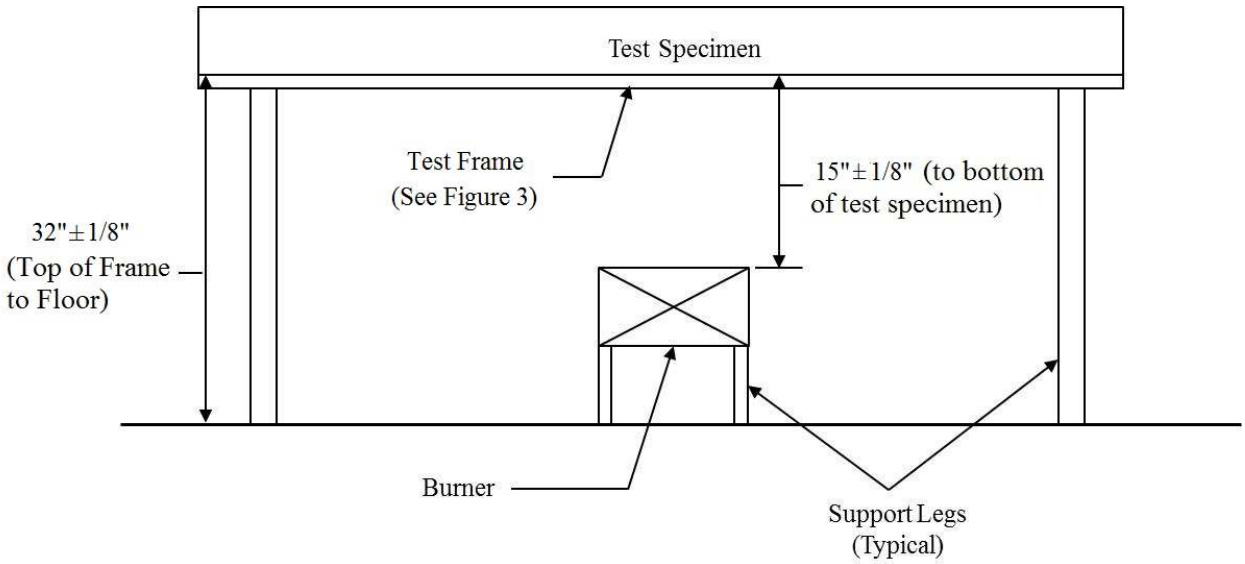
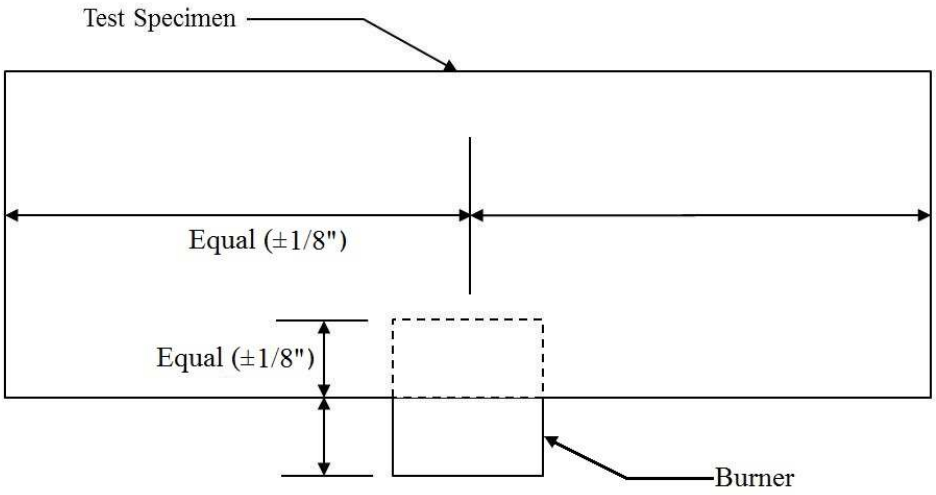


FIGURE 3. Gas burner.

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ELEVATION VIEW OF TEST SPECIMEN & BURNER



PLAN VIEW OF TEST SPECIMEN & BURNER

FIGURE 4. Test specimen and burner layout.

4.7.4.5.2 Average specific extinction area. Average specific extinction area (ft^2/lb) is defined as the total smoke released (ft^2) divided by the total mass loss (lb).

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4.7.5 Bend fatigue test.

4.7.5.1 Specimen. The specimen shall be the full specified product thickness by 4±¼ inches wide by at least four times its thickness in length.

4.7.5.2 Apparatus. The following apparatus shall be used:

a. Set of two parallel metal compression plates as shown on figure 5.

b. Aging chamber in accordance with ASTM D573.

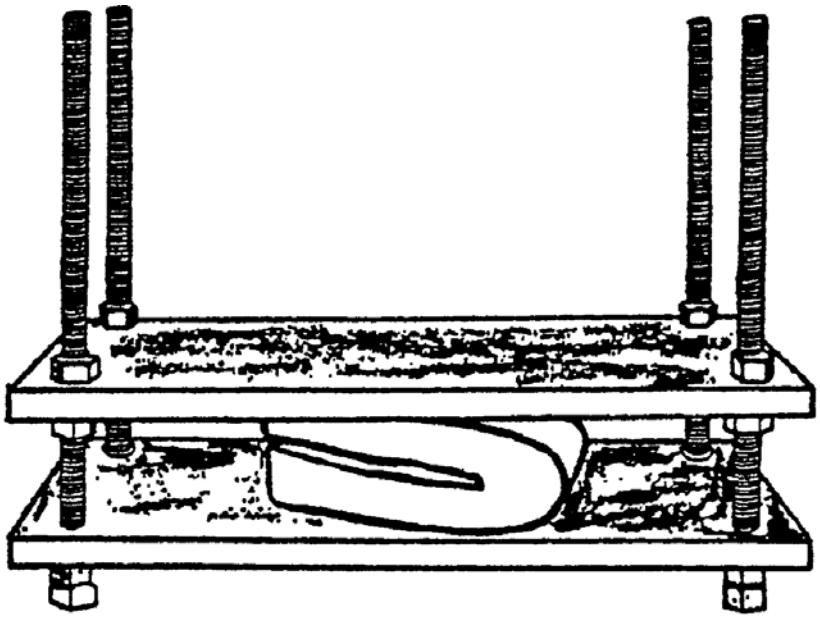


FIGURE 5. Specimen positioned in test apparatus for fatigue test.

4.7.5.3 Procedure. The specimen shall be folded crosswise and inserted between the two compression plates, skin side out, if there is more skin on one side. The opening between the compression plates shall be adjusted to two times the thickness of the unfolded specimen. The specimen shall be examined visually without magnification for cracks in the outer folded edge. The assembly shall be aged for 22±¼ hours at 158±4 °F. After aging, the specimen shall again be examined visually for cracks.

4.7.6 Tensile strength and ultimate elongation. Three test specimens, each 1±0.0625 by 6±0.0625 by 0.50±0.0625 inch, shall be cut from the top portion of the material. The length of the test specimens shall be parallel to the length of the material and the cut edges shall be perpendicular to the top surface and free of ragged edges. The tensile strength and ultimate elongation of the test specimens shall then be determined in accordance with the tensile test of ASTM D3574.

4.7.7 Cemented seam strength.

4.7.7.1 Specimen. The specimen shall be of any convenient length and shall have a cross-section of ½ by 1 inch. The cemented seam shall be across this cross-section, duplicating production as nearly as practical.

4.7.7.2 Apparatus. The following apparatus shall be used:

a. Testing machine in accordance with ASTM D3574.

b. Aging chamber in accordance with ASTM D573.

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4.7.7.3 Procedure. The specimen shall be tested perpendicular to the cemented seams in the testing machine. The elapsed time before testing the cemented seams shall be not less than 24 hours at 70±5 °F after cementing and not less than 70±½ hours at 158±4 °F after aging.

4.7.8 Odor test. The odor test shall be conducted in accordance with ASTM C739.

4.7.9 Dynamic fatigue. Material shall be tested in accordance with ASTM D3574 Test I_3, Procedure B.

4.8 Off-gassing. The finished foam shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 3.9 and 6.4). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA-approved test facility (see 3.9). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (3.9). Off-gas testing shall only be required once unless changes are made to the formulation, composition, or manufacturing process.

4.9 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.10.1 and 3.10.2, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the finished foam using data provided by the manufacturer/distributor to the NMCPHC (see 3.10.1 and 6.6).

4.10 Material. The material requirements of 3.2 shall be verified via a certificate of conformance as specified (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use in cushioning applications, such as upholstered furniture, and for packaging applications for military use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Condition required (see 1.2).
c. Whether first article testing is required (see 3.1 and 4.2).
d. Certificate of conformance (see 3.2 and 4.10).
e. Form of material required (sheets or shapes) (see 3.3.1).
f. Dimensions required (see 3.5).
g. Requirements for fire gas toxicity (see 3.6.4.3, 4.7.4.4, and 6.5).
h. If additional marking is required (see 3.7).
i. Whether the material is to be used on board a submarine (see 3.9, 4.8, and 6.4).
j. Inspection conditions, if other than as specified (see 4.6).
k. Additional fire testing provisions (see 4.7.4.1 and 6.7).
l. Requirements for an HHA (see 4.9 and 6.6).
m. Packaging requirements (see 5.1).

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6.3 Design information. Foam materials conforming to this specification were formerly designated in previous versions of this specification as Type II (uncored), Class 5.

6.4 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.5 Fire gas toxicity evaluation. NAVSEA recommends inclusion of the fire gas toxicity evaluation (see 3.6.4.3 and 4.7.4.4). Measurement of the concentration of each toxic gas should be as prescribed in 4.7.4.4 and provided in the first article test report (see 6.2). The Navy will use the data provided for evaluation purposes.

6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.7 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. NAVSEA will provide a letter that indicates the approved laboratories which are accredited to ISO/IEC 17025 and which details how to obtain and maintain accreditation (see 6.2).

6.8 Subject term (key word) listing.

Cushioning

Packaging, flame resistant

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MR
Navy – SH
DLA – GS

Preparing activity:
Navy – SH
(Project 9320-2014-001)

Review activities:
Army – AR, EA, MI
Navy – EC, MC
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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